Unit – II

Cloud Computing
(Course Code: 4360709)

Cloud Computing Architecture & Deployment Model

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Vision Mission

Vision of Computer Department

Develop globally competent Computer Engineering Professionals who achieve excellence in an environment conducive for technical knowledge, skills, moral and ethical values with a focus to serve the society.

Mission of Computer Department

M1 : To provide state of the art infrastructure and facilities for imparting quality education and computer engineering skills for societal benefit.

M2 : Adopt industry oriented curriculum with an exposure to technologies for building systems & application in computer engineering.

M3 : To provide quality technical professional as per the industry and societal needs, encourage entrepreneurship, nurture innovation and life skills in consonance with latest interdisciplinary trends.

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Unit-2: Cloud Service Model & service providers

CO2 - B) To compare the Deployment models with reference to Cloud Computing

2.1 Describe cloud computing architecture

- 2.1.1 Introduction Cloud Computing Architecture
- 2.1.2 Components of Cloud Computing Architecture

2.2 Describe various cloud deployment model

- 2.2.1 Types of Cloud Deployment Model

2.3 Compare and contrast different cloud deployment models, including public, private, hybrid, and community clouds.

- 2.3.1 Key drivers to adopting the cloud
- 2.3.2 The impact of Cloud computing on Users
- 2.3.3 A Comparative Analysis of Cloud Deployment Models

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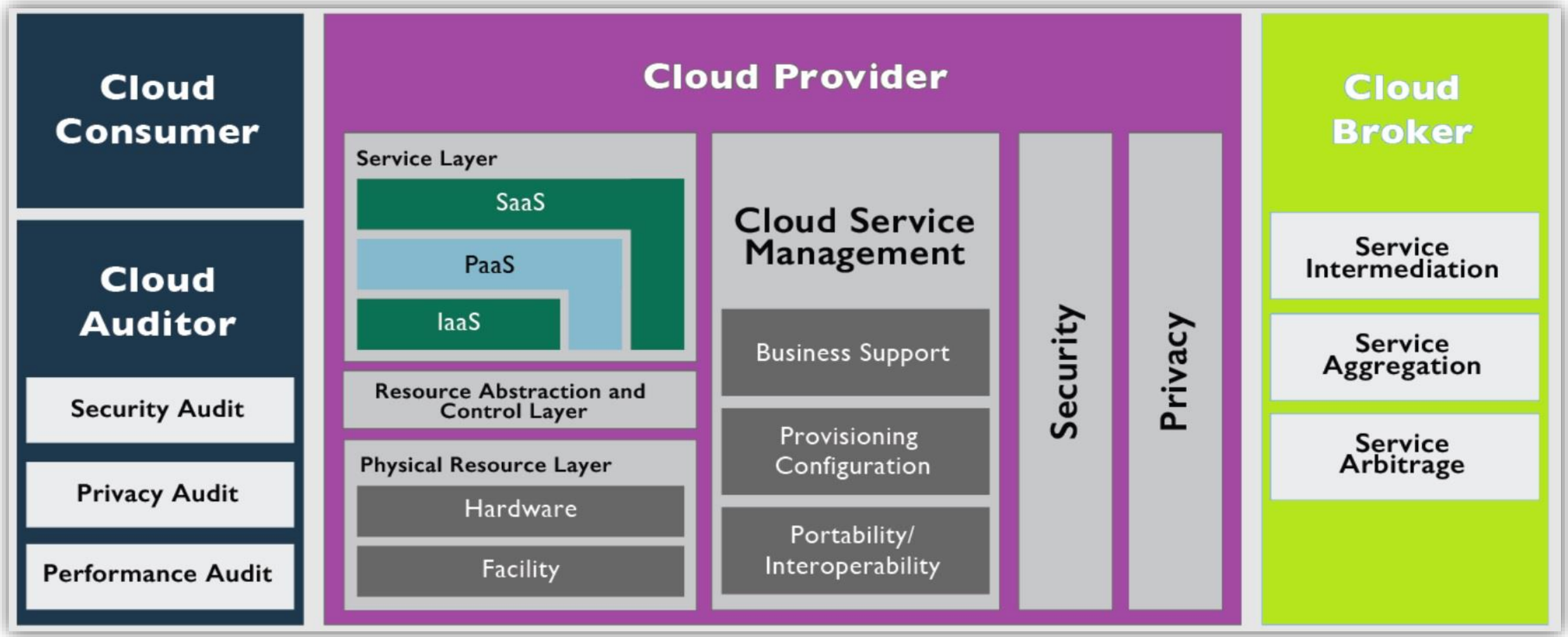
Cloud Architecture

NIST Architecture of Cloud



Cont…

- **Cloud Consumer**: An individual or organization that engages in a business partnership with Cloud Providers and utilizes their services.
- **Cloud Provider**: A person, organization, or entity that is accountable for offering services to individuals or entities who are interested in them.
- **Cloud Auditor**: A party capable of conducting an impartial evaluation of cloud services, as well as assessing the performance, security, and operations of the cloud implementation.
- **Cloud Broker**: An entity responsible for managing the utilization, performance, and delivery of cloud services. Additionally, they act as mediators between Cloud Providers and Cloud Consumers, negotiating relationships between the two parties.
- **Cloud Carrier**: An intermediary that facilitates the connectivity and transportation of cloud services from Cloud Providers to Cloud Consumers.

Cloud Architecture Best Practices

- **Up-front Planning**: Prioritize up-front planning to understand capacity needs and avoid unexpected production glitches. Continuously test performance during the architecture design phase.

- **Security First**: Implement robust security measures to protect against unauthorized access. Safeguard all layers of the cloud infrastructure through data encryption, patch management, and strict security policies. Consider adopting zero-trust security models for enhanced protection in hybrid and multi-cloud environments.

- **Disaster Recovery Readiness**: Automate recovery processes to minimize costly downtime and ensure swift recovery from service disruptions. Implement monitoring mechanisms to track capacity and utilize redundant networks for a highly available architecture.

- **Maximize Performance**: Optimize computing resources by continuously monitoring business demands and technological requirements. Efficiently manage and allocate resources to maintain optimal performance levels.

- **Cost Optimization**: Leverage automated processes, managed service providers, and utilization tracking to reduce unnecessary cloud computing expenses. Regularly review and optimize resource allocation to ensure cost-effectiveness.

Conclusion

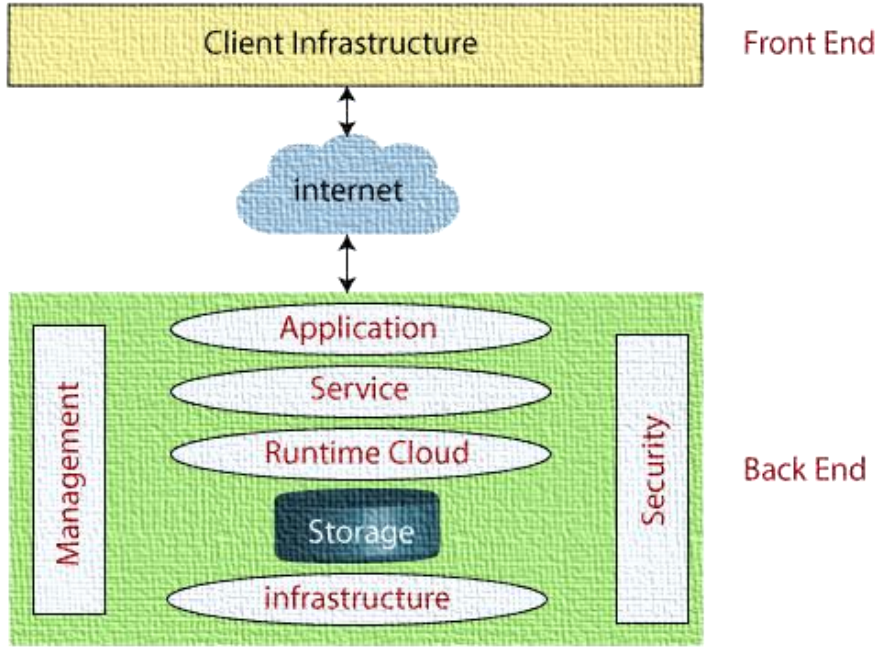
- Cloud computing enables the delivery of computing services over the internet, eliminating the need for local infrastructure.
- Key characteristics of cloud computing include on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
- Deployment models include public, private, and hybrid clouds, offering flexibility and data sharing options.
- Service models such as IaaS, PaaS, and SaaS provide virtualized computing resources, platform for application development, and software delivery respectively.
- Cloud computing offers benefits like cost savings, scalability, flexibility, reliability, and enhanced collaboration.
- Cloud computing presents a transformative approach to computing, offering cost-effective solutions, scalability, and flexibility for organizations across various industries. By leveraging the benefits of cloud computing and aligning with the appropriate deployment and service models, organizations can drive innovation, improve efficiency, and adapt to evolving business needs.

Remember

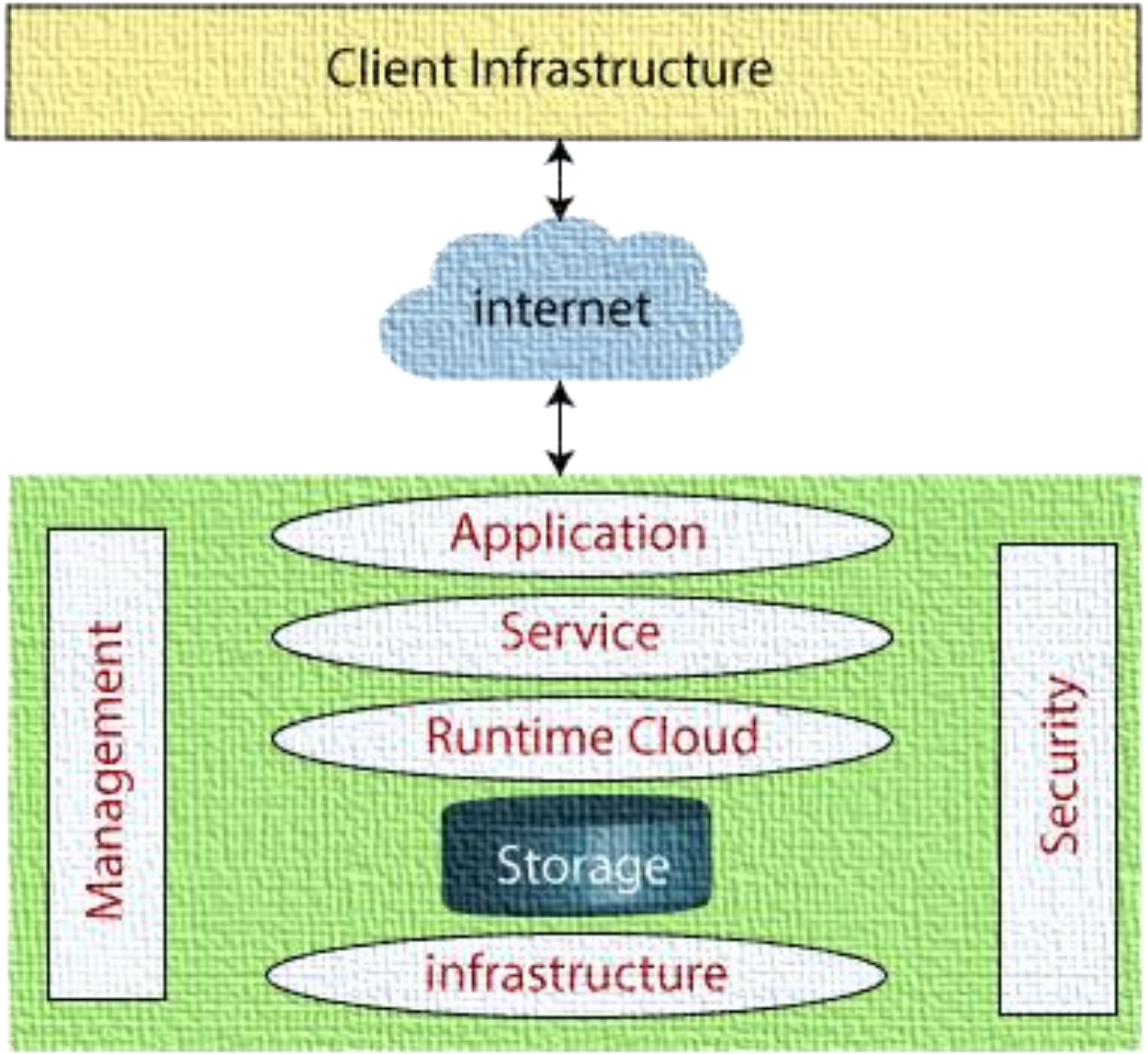
- Understand your specific needs and requirements when selecting a cloud deployment and service model.
- Prioritize security measures such as data encryption and access controls to protect against unauthorized access.
- Plan for disaster recovery and high availability to ensure business continuity.
- Continuously monitor and optimize resource utilization for optimal performance and cost savings.
- Regularly assess and adapt your cloud architecture to evolving business needs and emerging technologies.

Cloud Architecture

- Cloud architecture refers to the design and structure of cloud computing environments, including the arrangement of components and the relationships between them.
- It involves various elements that work together to deliver cloud services and ensure reliability, scalability, and security.



Cloud Architecture



Front End

Back End

Parts of Cloud computing architecture

1) **Front End:** Frontend of the cloud architecture refers to the client side of cloud computing system. Means it contains all the user interfaces and applications which are used by the client to access the cloud computing services/resources. For example, use of a web browser to access the cloud platform.

Client Infrastructure – Client Infrastructure is a part of the frontend component. It contains the applications and user interfaces which are required to access the cloud platform.

In other words, it provides a GUI(Graphical User Interface) to interact with the cloud which includes **web browsers (including Chrome, Firefox, internet explorer, etc.), thin & fat clients, tablets, and mobile devices**.

Backend :

Backend refers to the cloud itself which is used by the service provider.

- It contains the **resources** as well as **manages** the resources and provides **security** mechanisms.
- Along with this, it includes huge storage, virtual applications, virtual machines, traffic control mechanisms, deployment models, etc.

Components of Cloud Computing Architecture

1. Client Infrastructure
2. Cloud access device
3. Application
4. Services
5. Runtime cloud
6. Storage
7. Infrastructure
8. Security

Components of Cloud Computing Architecture

1. **Client Infrastructure**: Cloud Infrastructure in backend refers to the hardware and software components of cloud like it includes servers, storage, network devices, virtualization software etc.

2. **Cloud access device**: It is the client end device used to access the cloud services over internet.

   i.e. Desktop computers, Laptops, Tablets, Smartphones.

3. **Application**: Application in backend refers to a software or platform to which client accesses. It provides the service in backend as per the client requirement.

Components of Cloud Computing Architecture

4. Services:

A Cloud Services manages that which type of service you access according to the client's requirement.

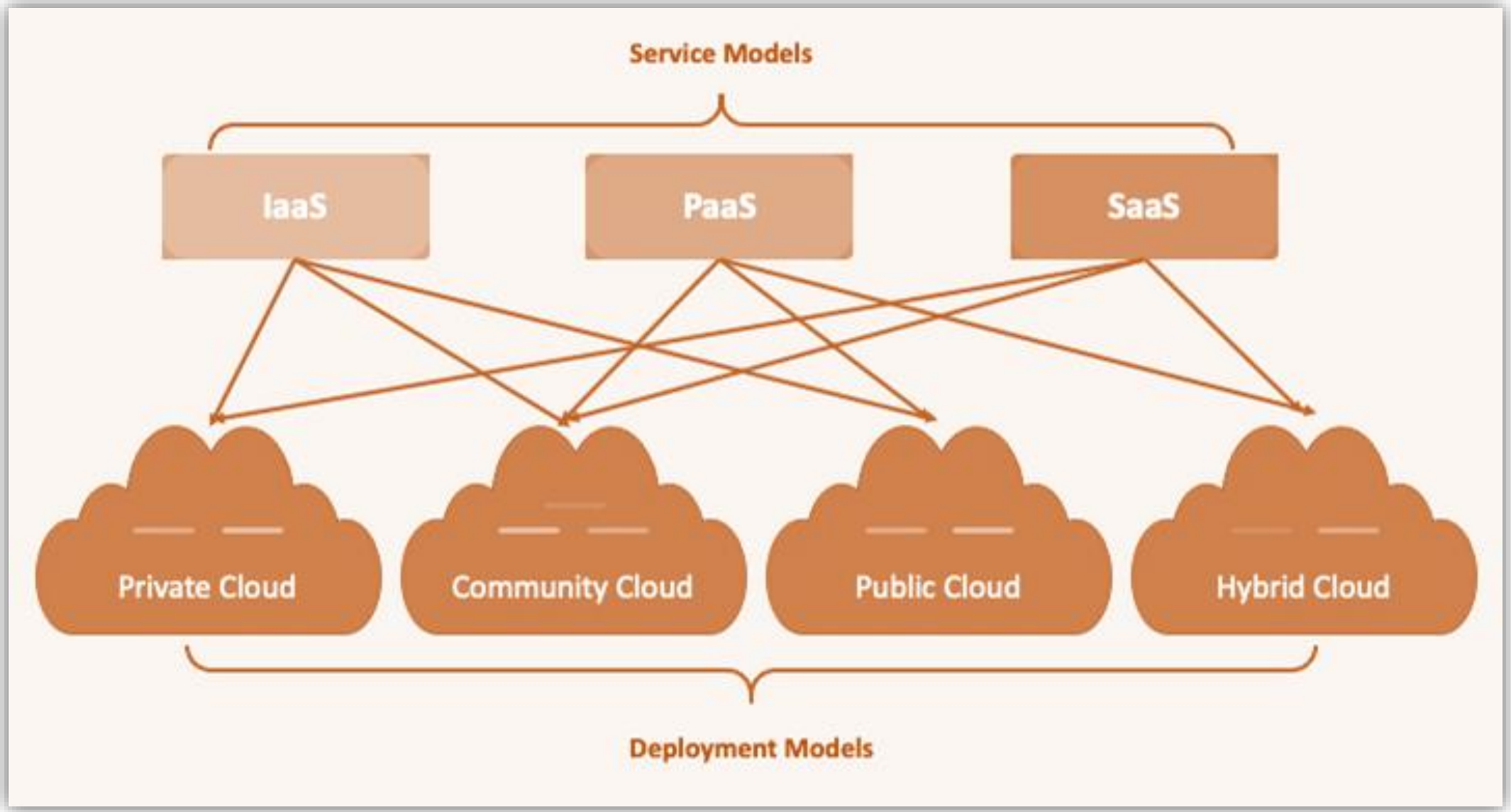
Cloud computing offers the following three type of services:

i. Software as a Service (SaaS) – It is also known as **cloud application services.**

Mostly, SaaS applications run directly through the web browser means we do not require to download and install these applications. Some important examples of SaaS are as below –

Example: Google Apps, Salesforce Dropbox, Slack, Hubspot, Cisco WebEx.

Cloud Deployment Models



Components of Cloud Computing Architecture

ii. Platform as a Service (PaaS) – It is also known as **cloud platform services**. It is quite similar to SaaS, but the difference is that PaaS provides a platform for software creation, but using SaaS, we can access software over the internet without the need of any platform.

Example: Windows Azure, Force.com, Magento Commerce Cloud, OpenShift.

iii. Infrastructure as a Service (IaaS) – It is also known as **cloud infrastructure services**. It is responsible for managing applications data, middleware, and runtime environments.

Example: Amazon Web Services (AWS) EC2, Google Compute Engine (GCE), Cisco Metapod.

Components of Cloud Computing Architecture

6. **Run Time Cloud**: Runtime Cloud provides the execution and runtime environment to the virtual machines.

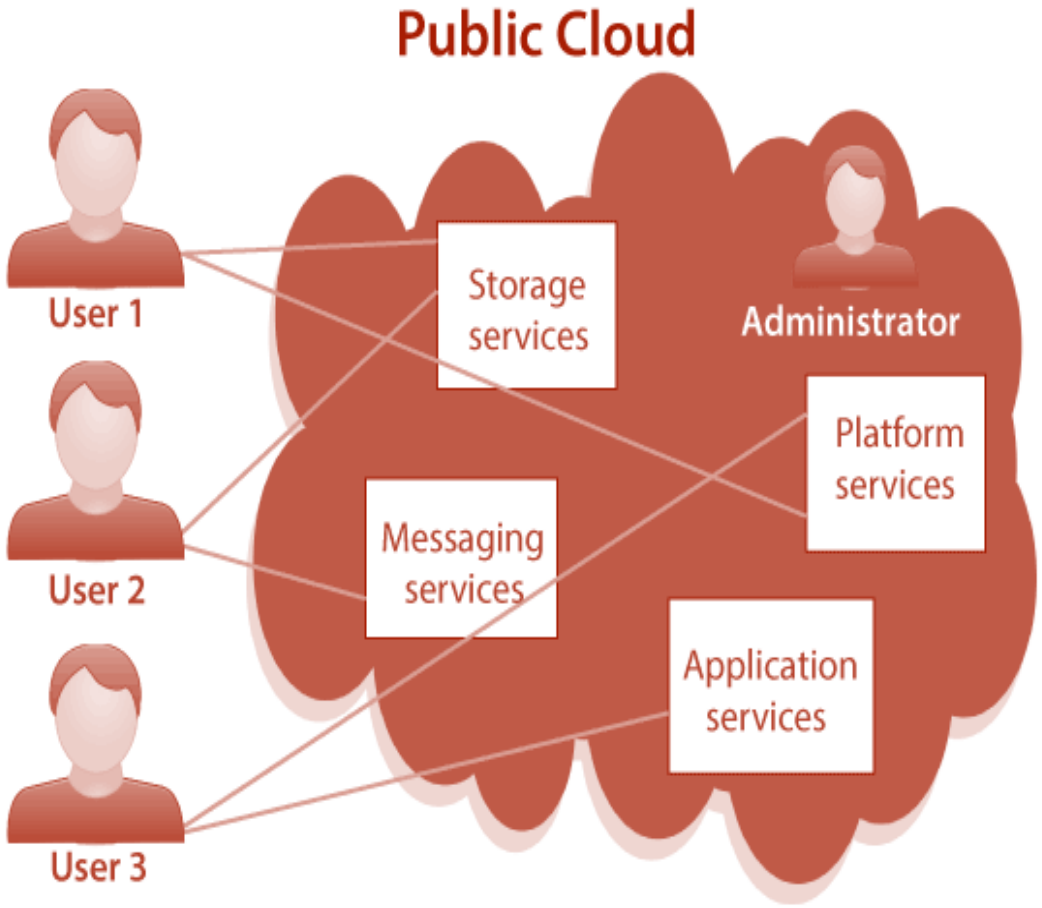
7. **Storage**: It provides a huge amount of storage capacity in the cloud to store and manage data.

8. **Infrastructure**: It provides services on the host level, application level, and network level. Cloud infrastructure includes hardware and software components such as servers, storage, network devices, virtualization software, and other storage resources that are needed to support the cloud computing model.

9. **Security**: Security is an in-built back end component of cloud computing. It implements a security mechanism in the back end.

Public Cloud

- Services are provided over a public network and available to anyone who wants to use them.
- It is a cost-effective option for businesses and individuals looking for scalability and flexibility.
- Public cloud providers, such as AWS, Azure, and GCP, offer a wide range of services accessible to the general public.



Public Cloud

- A public cloud is probably the most **commonly understood** cloud computing option.
- This is where all the services and supporting infrastructure are managed **off-site over the Internet** and **shared across multiple** users (or tenants).
- A good example of a public cloud at the individual consumer level is a streaming service like **Netflix or Hulu**.
- Users subscribe to the service through an individual account but **access the same services across the platform** through the Internet.

Public Cloud

- The advantage of using a public cloud is the increased efficiency and **subsequent cost-effectiveness** from shared resources.
- Public clouds are typically cheaper than private and hybrid cloud solutions (as well as traditional on-premise computing) because they rely on economies of scale.
- Users don’t have to pay for services they aren’t using and don’t have to worry about managing and maintaining the physical infrastructure.

Private Cloud

- Infrastructure is dedicated to a single organization and may be located on-premises or off-premises.
- Private cloud environments are designed to meet specific security, compliance, or performance requirements.
- They offer enhanced control, customization, and privacy but require significant upfront investment.

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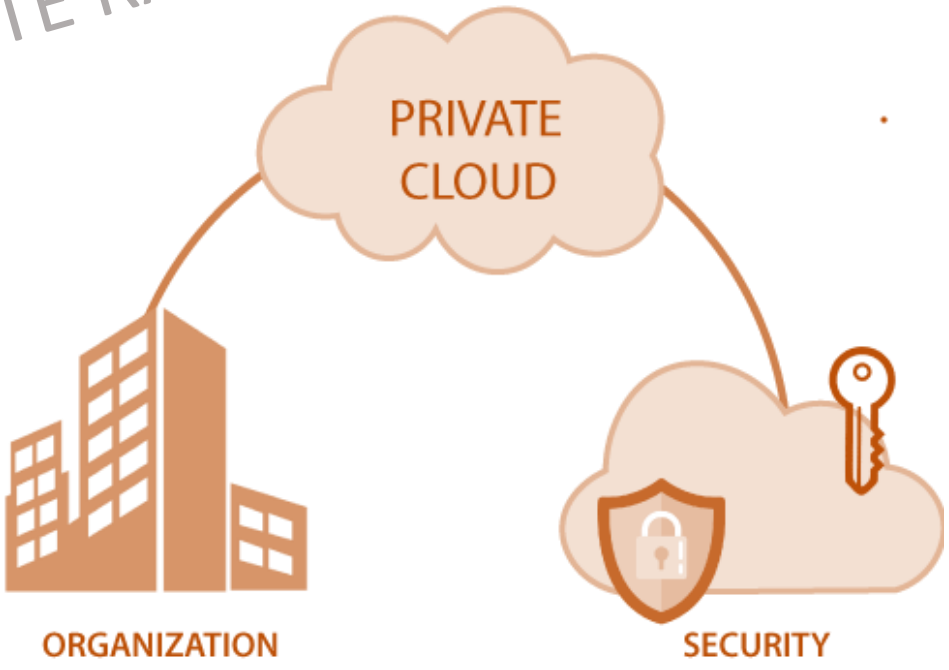


Private Cloud

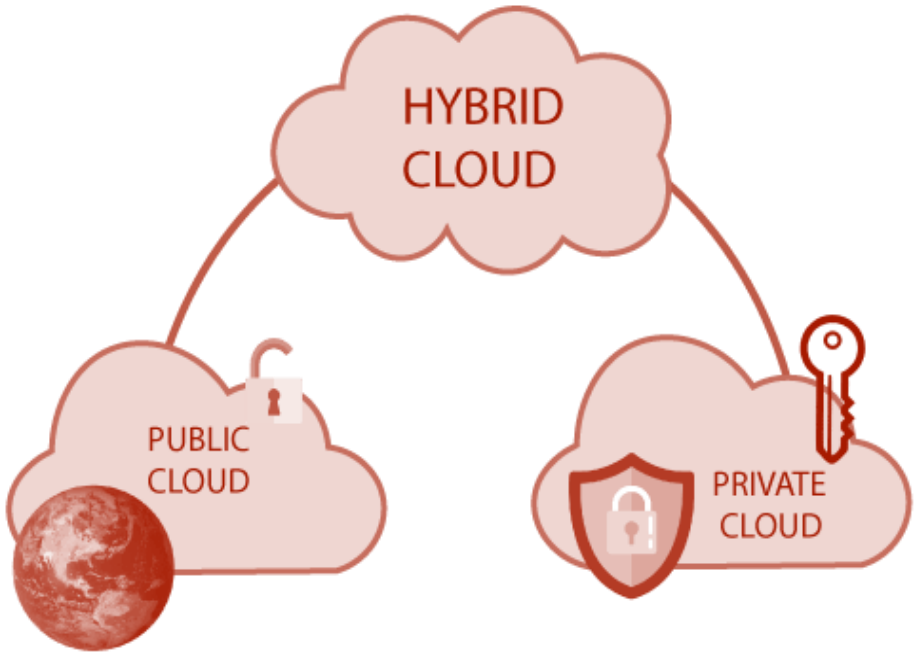
- A private cloud **provides IT services through the Internet or a private network to select users**, rather than to the general public.
- Instead of having **multiple tenants**, like a public cloud does, a private cloud typically has only one tenant.
- All the data is **protected** behind a firewall.
- This is a popular choice for many businesses who want the agility of the cloud with **greater customization and security.**

Private Cloud

- Private clouds can **reside on-site or off-site**.
- The distinguishing feature is the single, private tenant who maintains greater control over the IT services.
- Private clouds are popular choices for organizations who have **high priorities on security and compliance.**

Hybrid Cloud

- Combines public and private cloud environments, allowing for flexibility and data sharing between the two.
- Organizations can leverage the benefits of both public and private clouds, ensuring optimal resource allocation.
- Hybrid cloud deployments enable workload portability and seamless integration between different environments.



Hybrid Cloud

- A hybrid cloud environment combines both **private and public cloud** elements to varying degrees.
- Despite operating independently, the clouds in a hybrid environment communicate over an **encrypted connection** and allow for the portability of data and applications.
- This is an increasingly popular cloud solution because it allows organizations **greater flexibility** to meet their IT needs.

Community Cloud

- Community cloud is a deployment model where infrastructure and services are shared among a specific community or group of organizations.
- It caters to the needs of a particular community, such as government agencies, educational institutions, or research organizations.
- Community cloud provides a cost-effective solution while addressing specific requirements and compliance standards of the community.

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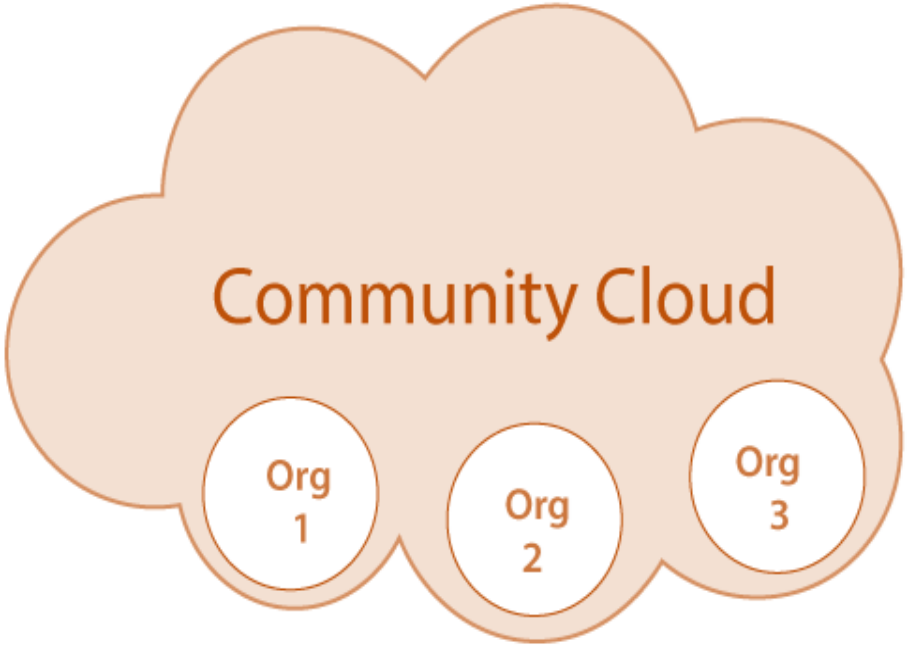
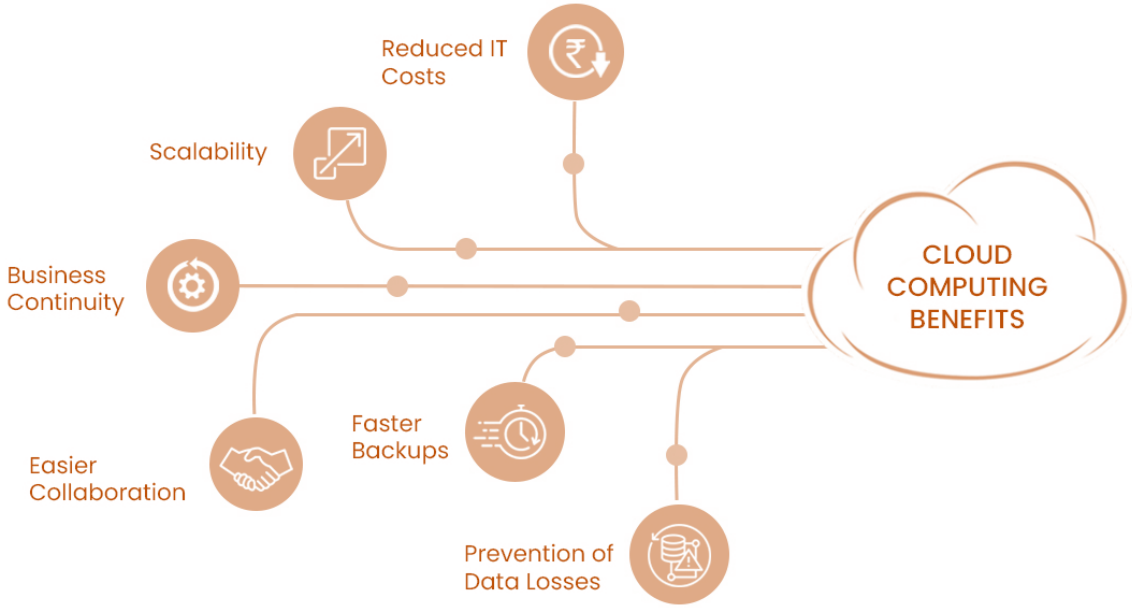
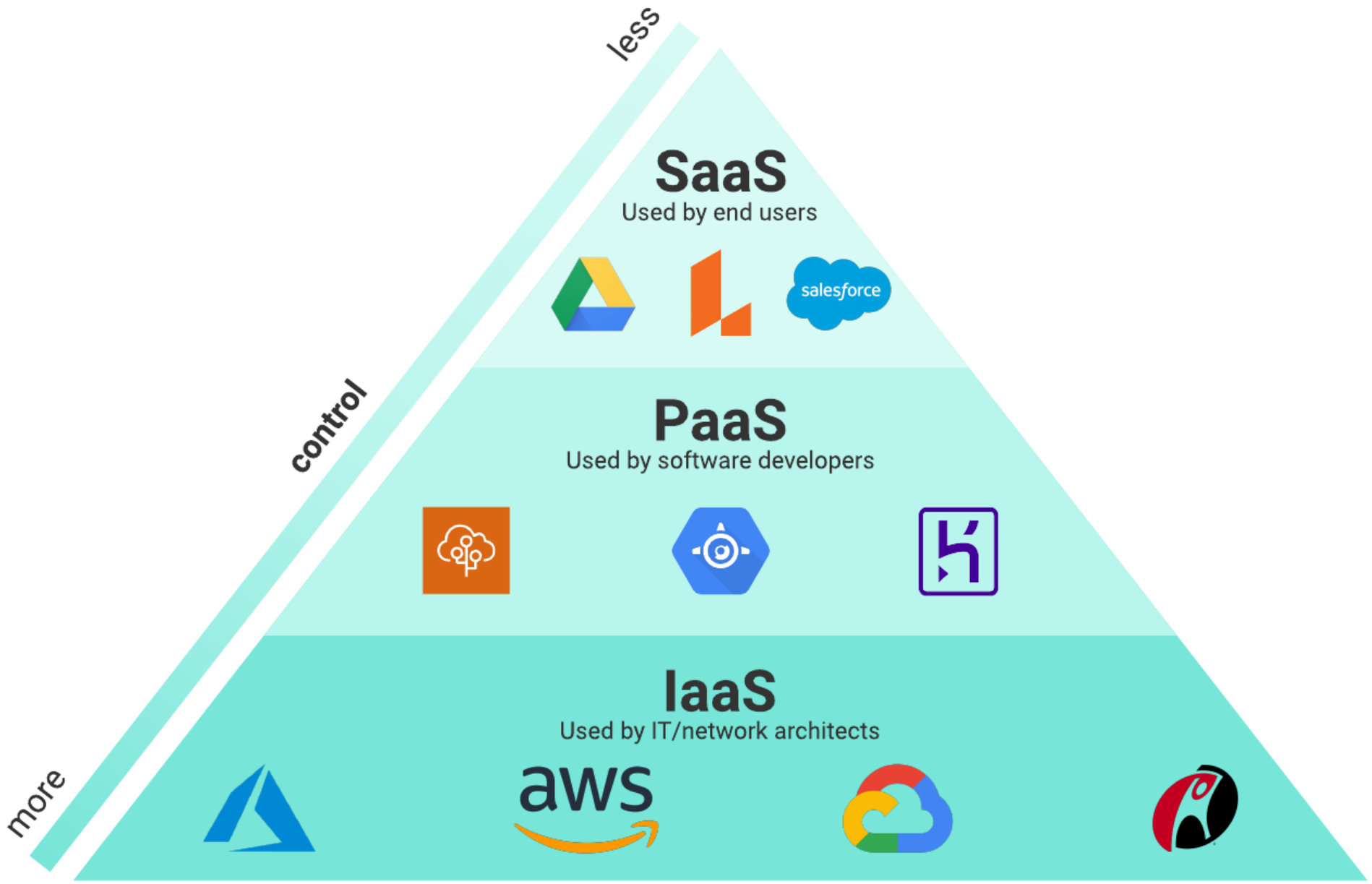


Benefits of Cloud

- **Cost Savings**: Pay for what you use, with no upfront infrastructure costs.
- **Scalability**: Easily scale resources up or down based on demand.
- **Flexibility**: Access resources and applications from anywhere with an internet connection.
- **Reliability**: Cloud providers typically offer high uptime and data redundancy.
- **Collaboration**: Enable seamless collaboration and data sharing among teams.




less
control
more
SaaS
Used by end users
salesforce
PaaS
Used by software developers
IaaS
Used by IT/network architects
aws

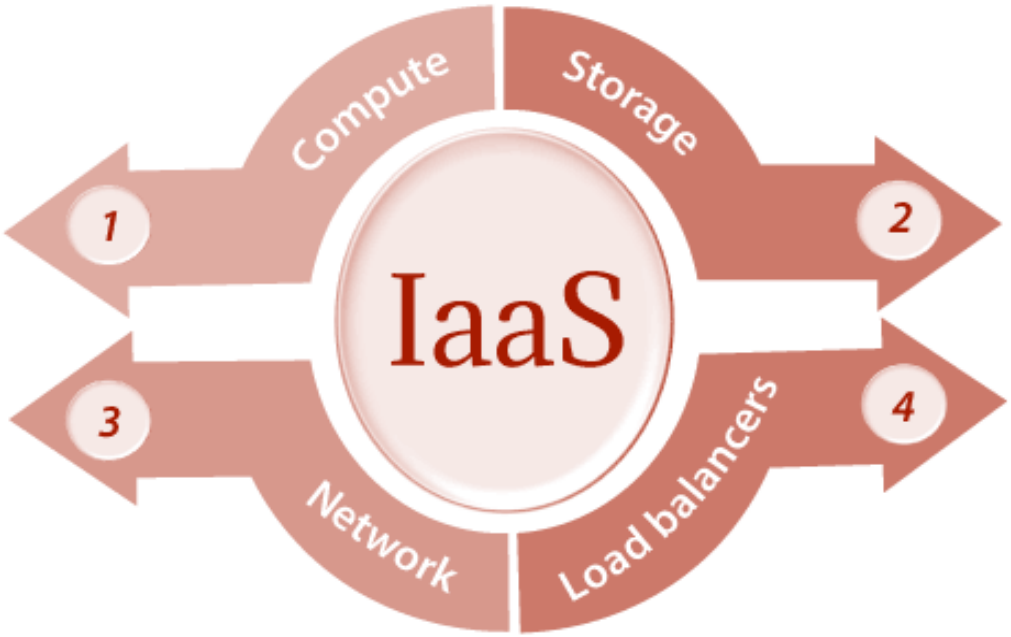
IaaS

- IaaS gives users access to storage, networking, servers, and other computing resources via the cloud. While the user is still responsible for managing their applications, data, middleware, etc., IaaS provides automated and scalable environments that provide a high degree of control and flexibility for the user.
- For example, many businesses use IaaS to support workload spikes during busy seasons (like holidays).
- Popular IaaS providers include:
- Amazon Web Services (AWS)
- Microsoft Azure
- Google Compute Engine (GCE), the IaaS component of Google Cloud Platform (GCP)

Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources over the internet. Users have control over the operating systems, storage, and networking components.

They can provision and manage virtual machines (VMs), storage, and networks according to their requirements. Examples of IaaS providers include AWS EC2, Azure Virtual Machines, and Google Compute Engine.

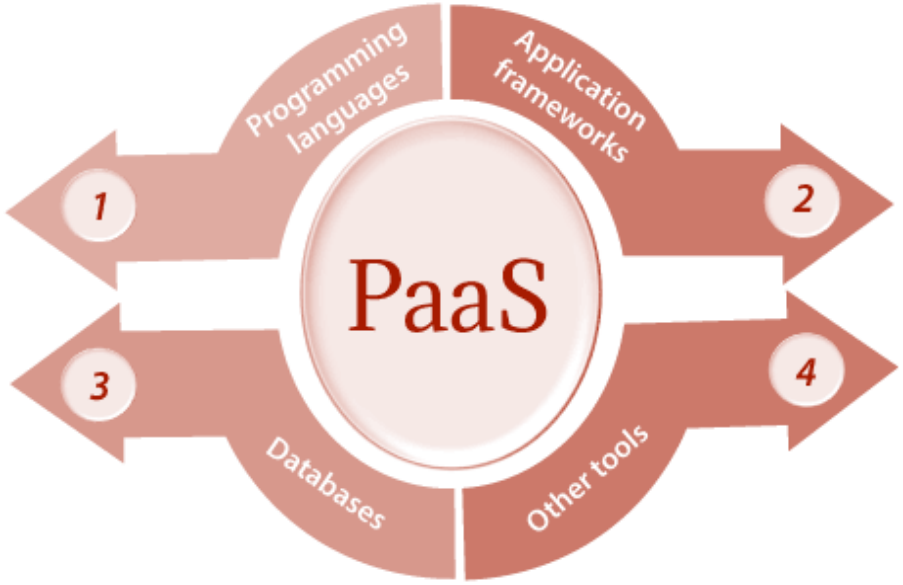


PaaS

- This service layer is primarily geared towards developers and operations professionals.
- Service providers rent out cloud-based platforms for users to develop and deliver applications. In other words, PaaS provides a framework that makes it easier and more efficient to build, customize, and deploy applications.

Platform as a Service (PaaS)

- PaaS offers a platform for developing, testing, and deploying applications.
- Users can focus on application development without worrying about infrastructure management.
- PaaS providers manage the underlying infrastructure, including servers, storage, and networking.
- Developers can leverage pre-configured environments, development frameworks, and deployment tools.
- Examples of PaaS providers include Heroku, Google App Engine, and AWS Elastic Beanstalk.



SaaS

- Cloud application services are the most well-known of the cloud service models.
- The software is hosted, packaged, and delivered by a third party through the Internet (typically on a browser-based interface). By delivering the software application over the Internet, enterprises can offload the costs of management and maintenance to the vendor(s).
- Popular SaaS options include email and customer relationship management software.

Software as a Service (SaaS)

- SaaS delivers software applications over the internet on a subscription basis.
- Users can access and use applications directly through a web browser or APIs.
- The provider hosts and manages the underlying infrastructure, application, and data.
- Users can typically customize certain aspects of the application to fit their needs.
- Examples of SaaS include Salesforce, Microsoft Office 365, and Google Workspace.

Key drivers to adopting the cloud

1) Security:

Security is one of the main reasons businesses are moving to the cloud. As cyber-attacks increase, companies want to bolster cyber security by moving to modern platforms.

Phishing and Malware attacks have also grown to be more sophisticated and are very difficult to combat. Cloud computing makes the business more secure against all these invading threats, and hence it is a key business driver for cloud adoption.

2) Cost Saving:

Hardware, storage, and network devices for an on-premises data center require a considerable investment. With the cloud, the capital expenditure gets reduced significantly. The pay per use cloud model enables one to pay only for what is used and thus saves a big chunk of ones investment which can be used in other productive pursuits of business expansion.

Key drivers to adopting the cloud

3) Efficiency:

Efficiency removes unnecessary steps to streamline the process. It increases productivity and gives a faster customer delivery approach.

4) Flexibility and Scalability:

Your computing services increase with your business growth. Maintaining them on your means increased costs and new infrastructure. But with the cloud, you can purchase more storage to meet your needs and have instant access to expanded capabilities. Cloud solutions are highly flexible and scalable.

Key drivers to adopting the cloud

5) Rapid recovery:

If you store your data on-premises, you need to have a replication of it on some other location, and if that is not done, then you have to replace the entire infrastructure if a disaster strikes. Both these things prove very costly.

But with cloud backup, your data is stored across several storage centers, and so even when a disaster strikes at one of the centers, it can be overcome quickly, using the other storage centers and backups.

6) Increased Convenience:

Cloud-based storage enables users to easily access the files they need, wherever they are stored. It becomes more convenient for the employees to use the technology and focus on other tasks leading to business growth.

Key drivers to adopting the cloud

7) Speed and Productivity:

It takes weeks or months to get new applications to run on traditional platforms. But in the cloud, it is a matter of hours. This reduces your time significantly and thus increases your productivity.

8) Strategic value:

Migrating to the cloud gives a competitive edge to businesses as they can give their clients and customers innovative technologies that are easily accessible. With the cloud, organizations can provide their customers with quick solutions and thus increasing business agility and customer satisfaction.

9) Multi-tenancy:

Cloud servers are architected so that several customers can share the underlying infrastructure resources without compromising their privacy and security.

10) Service and Innovation:

With the cloud, you can use many cloud services and application programming interfaces (APIs) and use flexible cloud tools and environments to build new and innovative applications and processes.

The impact of Cloud computing on Users

1. **Accessibility**: Users can access their data and applications from anywhere with an internet connection. This has changed how people work, communicate, and collaborate.
2. **Cost savings:** Users don't need to buy and maintain expensive hardware and software.
3. **Scalability and flexibility:** Users can choose which services to use and pay for what they use. They can also switch from one cloud to another.
4. **Increased productivity:** Employees can focus on their core tasks instead of technical issues.
5. **Tool selection:** Users can choose from a menu of prebuilt tools and features to build a solution that fits their needs.
6. **Security features**: Virtual private cloud, encryption, and API keys help keep data secure.

Comparative Analysis of Cloud Deployment Models

Factors	Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Initial Setup	Easy	Complex, requires a professional team to setup	Complex, requires a professional team to setup	Complex, requires a professional team to setup
Scalability and Flexibility	High	High	Fixed	High
Cost-Comparison	Cost-Effective	Costly	Distributed cost among members	Between public and private cloud
Reliability	Low	Low	High	High
Data Security	Low	High	High	High
Data Privacy	Low	High	High	High

FACTOR	Public	Private	Community	Hybrid
Ease of use	Very easy to uSse	Complex and requires an in-house team	Relatively easy to use as members help solve problems and establish protocols	**Difficult to use if the system was not set up properly**
Data control	Low, the provider has all control	Very high as yo-u own the system	High (if members collaborate)	**Very high (with the right setup)**
Reliability	Prone to failures and outages	High (with the right team)	Depends on the community	**High (with the right setup)**
Scalability	**Low, most providers offer limited resources**	**Very high as there are no other system tenants**	**Fixed capacity limits scalability**	**High (with the right setup)**

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	Public	Private	Community	Hybrid
Setup flexibility	Little to no flexibility, service providers usually offer only predefined setups	Very flexible	Little flexibility, setups are usually predefined to an extent	**Very flexible**
Cost	Very Inexpensive	Very expensive	Members share the costs	**Cheaper than a private model, pricier than a public one**
Demand for in-house hardware	**No**	**In-house hardware is not a must but is preferable**	**No**	**In-house hardware is not a must but is preferable**

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References

- Prof. Hiral B. Patel, Prof. Nirali Kansara, Cloud Computing Deployment Models: A Comparative Study, International Journal of Innovative Research in Computer Science & Technology (IJIRCST) ISSN: 2347-5552, Volume-9, Issue-2, March 2021
- https://www.geeksforgeeks.org/cloud-deployment-models
- https://www.javatpoint.com/cloud-computing-architecture

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